Word Problems & Reasoning

ANSWER KEY

Section 1: Age Word Problems

1 A father is currently 4 times as old as his son. In 10 years, the father will be only 2 times as old as his son will be then. How old is the father now?

a. 20 **years old ✓**

b. 24 years old

c. 28 years old

d. 16 years old

Let the son's age be s, father's age $f = 4s$. In 10 years:
$4s + 10 = 2(s + 10) \Rightarrow 4s + 10 = 2s + 20 \Rightarrow 2s = 10 \Rightarrow s = 5$. Father's age $= 4 \times 5 = 20$.

2 The sum of the ages of Omar and Zaid is 35 years. If Omar is 5 years older than Zaid, what is Zaid's age?

a. 15 **years old ✓**

b. 18 years old

c. 20 years old

d. 12 years old

Let Zaid's age be z. Omar's age $= z + 5$. Sum: $z + (z + 5) = 35 \Rightarrow 2z = 30 \Rightarrow z = 15$.

3 Three years ago, Sarah's age was twice her brother's age. If Sarah is now 15 years old, how old is her brother now?

a. 9 **years old ✓**

b. 8 years old

c. 10 years old

d. 7 years old

Three years ago, Sarah was $15 - 3 = 12$ years old. Her brother was $\frac{12}{2} = 6$ years old. Therefore, his current age is $6 + 3 = 9$ years old.

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Section 2: Clock Angles & Time Intervals

4 What is the measure of the smaller angle between the hour hand and the minute hand of an analog clock at $4:00$?

a. 120° ✓

b. 100°

c. 90°

d. 110°

Each hour on a clock represents $\frac{360^\circ}{12} = 30^\circ$. At $4:00$, the hands are 4 hours apart: $4 \times 30^\circ = 120^\circ$.

5 What is the angle between the hour and minute hands at $2:30$?

a. 105° ✓

b. 90°

c. 115°

d. 100°

Formula: $|30H - 5.5M| = |30(2) - 5.5(30)| = |60 - 165| = |-105| = 105^\circ$.

6 A clock gains 5 minutes every hour. If it is set correctly at $12:00$ PM, what time will it display at $6:00$ PM on the same day?

a. $6:30$ PM ✓

b. $6:25$ PM

c. $6:35$ PM

d. $6:20$ PM

Elapsed time $= 6$ hours. Total gain $= 6 \times 5 = 30$ minutes. Displayed time $= 6:30$ PM.

Section 3: Profit, Loss & Commercial Math

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7 A merchant bought a device for 800 SAR and sold it at a 25% profit. What was the selling price?

a. 1000 **SAR** ✓

b. 950 SAR

c. 1050 SAR

d. 900 SAR

Profit $= 0.25 \times 800 = 200$ SAR. Selling price $= 800 + 200 = 1000$ SAR.

8 A store sells an item for 180 SAR after applying a 10% discount. What was the original price before the discount?

a. 200 **SAR** ✓

b. 190 SAR

c. 210 SAR

d. 220 SAR

Let original price be P. $0.90P = 180 \Rightarrow P = \frac{180}{0.90} = 200$ SAR.

9 If a car was purchased for $50,000$ SAR and sold for $42,500$ SAR, what was the percentage loss?

a. 15% ✓

b. 12.5%

c. 17.5%

d. 10%

Loss $= 50,000 - 42,500 = 7,500$ SAR. Percentage loss $= \frac{7500}{50000} \times 100\% = 15\%$.

Section 4: Logic, Queuing & Overlap Reasoning

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10 In a line of people, Tariq is 8th from the front and 12th from the back. How many people are in the line?

a. 19 ✓

b. 20

c. 21

d. 18

Total in line $= \text{Front rank} + \text{Back rank} - 1 = 8 + 12 - 1 = 19$.

11 In a class of 30 students, 18 play football, 15 play basketball, and 5 play neither sport. How many students play both sports?

a. 8 ✓

b. 7

c. 9

d. 6

Students playing at least one sport $= 30 - 5 = 25$. By inclusion-exclusion:
$18 + 15 - \text{Both} = 25 \Rightarrow 33 - \text{Both} = 25 \Rightarrow \text{Both} = 8$.

12 A box contains 10 black socks and 10 white socks mixed together. What is the minimum number of socks one must draw in the dark to guarantee at least one matching pair?

a. 3 ✓

b. 2

c. 4

d. 11

By the pigeonhole principle: There are 2 colors. Drawing $2 + 1 = 3$ socks guarantees that at least two must share the same color.

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